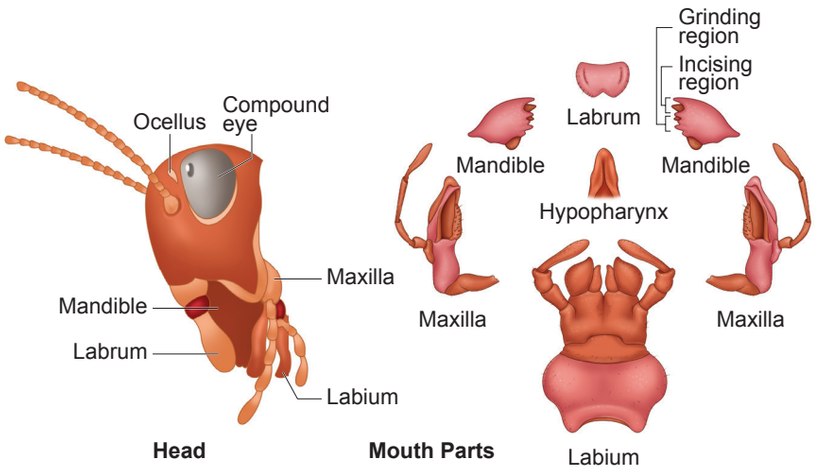


Figure 4.9 *Periplaneta americana*

antennae and appendages around the mouth. Antennae have sensory receptors that help in monitoring the environment. The appendages form the mouth parts which are of biting and chewing type (**Mandibulate** or **Orthopterus type**).

The mouth parts consist of a **labrum** (upper lip), a pair of **mandibles**, a pair of **maxillae**, a **labium** (lower lip) and a **hypopharynx** (tongue) or **lingua** (Figure 4.9).

The thorax consists of three segments – **Prothorax**, **Mesothorax** and **Metathorax**. The prothoracic segment is the largest. The head is connected with thorax by a short extension of the prothorax called as the **neck** or **cervicum**. Each thoracic segment bears a pair of walking legs. Due to the presence of three pairs of walking legs it is also called **hexapoda** (hexa-six, poda-feet) All the three pairs of walking legs are similar and each leg consists of five segments – **coxa** (large), **trochanter** (small), **femur** (long and broad), **tibia** (long and thick) and **tarsus**. The last segment of the leg - tarsus has five movable joints or **podomeres** or **tarsomeres**. Cockroach has two pairs of wings, the first pair arises from mesothorax and protects the hind wings when at rest, and is called **elytra** or **tegmina**. The second pair of wings arises from the metathorax and are used in flight. The abdomen in both male and female consists of 10 segments. Each segment is covered by the dorsal tergum, the ventral sternum and between them a narrow membranous pleuron on each side. In females, the 7th sternum is boat shaped and together with the 8th and 9th sterna forms a **brood** or **genital pouch** whose anterior parts contains female gonopore, spermathecal pores, collaterial glands and posterior parts constitutes the oothecal chamber in which the cocoons are formed. In males, the genital pouch lies at the hind end of the abdomen bound dorsally by 9th and 10th terga and ventrally by the 9th sternum. It contains the dorsal anus and ventral male genital pore. In both the sexes, genital apertures are surrounded by sclerites called **gonapophysis**. Male bears a pair of short and slender **anal styles** in the 9th sternum which are absent in the female. In both sexes, the 10th segment bears a pair of jointed filamentous structures called **anal cerci** and bears a sense organ that is receptive to vibrations in air and land. The 7th sternum of male has a pair of large and oval apical lobes or gynovalvular plates which form a keel like structure which distinguishes the male from the female.



One of the fastest moving land insects is the cockroach. They can move as fast as 5.4 Km per hour.

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